Unit 4 Pre-Test

Angle Relationships & Triangles

Standard assessed: NC.8.G.5

Name: _____ **Date:** _____ **Period:** _____

SCORE 3.0 Basic Understanding

NC.8.G.5 Understand angle relationships at a point and on a line.

• *I can find missing angles using supplementary, complementary, vertical, and linear-pair relationships.*

Match each vocabulary term to its definition, then find each missing angle.

D supplementary angles ✓	**A.** Two angles whose measures add to 90°.
A complementary angles ✓	**B.** Opposite angles formed by two intersecting lines; they are equal.
B vertical angles ✓	**C.** A line that crosses two or more other lines.
C transversal ✓	**D.** Two angles whose measures add to 180°.
E corresponding angles ✓	**E.** Angles in the same corner at each crossing. They are equal when the lines are parallel.
F exterior angle ✓	**F.** The angle formed outside a triangle when one side is extended.

Angle facts — reference

Straight line = **180°** · Triangle angles sum to **180°** · Right angle = **90°**

Exterior angle of a triangle = sum of the two remote interior angles.

1. *Supplementary*

Two angles are supplementary. One is 125°. Find the other.

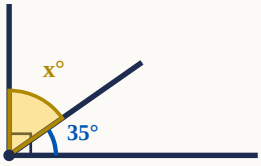
2. *Complementary*

Two angles are complementary. One is 35°. Find the other.



x =

answer: 55°

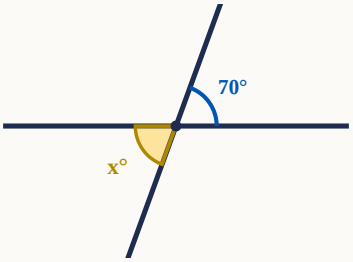


x =

answer: 55°

3. *Vertical*

Two lines cross. One angle is 70°. Find its vertical angle.

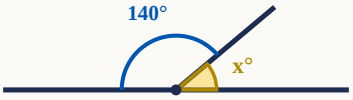


x =

answer: 70°

4. *Linear pair*

Angles on a straight line are 140° and x. Find x.

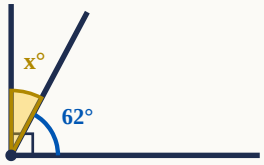


x =

answer: 40°

5. *Complementary*

Two angles are complementary. One is 62°. Find the other.

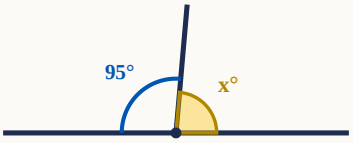


x =

answer: 28°

6. *Linear pair*

Two angles form a linear pair. One is 95°. Find the other.



x =

answer: 85°

SCORE *4.0* **Target Content**

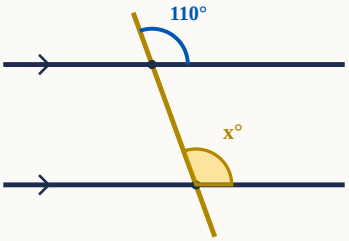
NC.8.G.5 Find angles formed when parallel lines are cut by a transversal.

• *I can use corresponding, alternate, and same-side interior angle relationships to find unknown angles.*

Find each unknown. Where a variable is given, solve for x.

7. *Corresponding*

Parallel lines cut by a transversal. Find the corresponding angle to 110°.

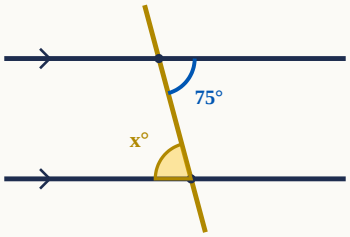


x = _____

answer: 110°

8. *Alt. interior*

Alternate interior angles; one is 75°. Find the other.

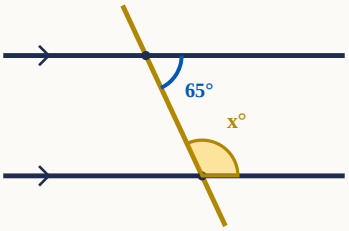


x = _____

answer: 75°

9. *Same-side interior*

Same-side interior angles; one is 65°. Find the other.

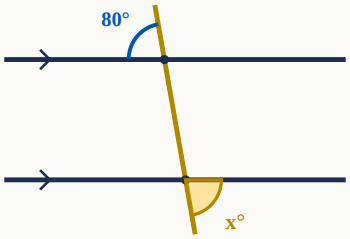


x = _____

answer: 115°

10. *Alt. exterior*

A transversal makes an 80° angle. Find its alternate exterior angle.



x = _____

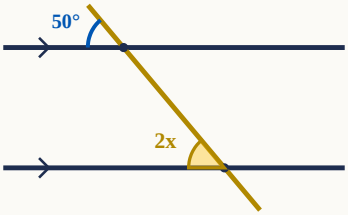
answer: 80°

11. *Solve*

Angles 2x and 50° are corresponding. Find x.

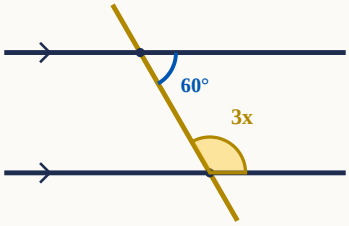
12. *Solve*

Same-side interior angles 3x and 60° add to 180°. Find x.



x = _____

answer: x = 25



x = _____

answer: x = 40

SCORE 5.0 Advanced Content

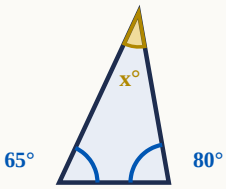
NC.8.G.5 Apply the triangle angle-sum and exterior-angle theorems.

• *I can use the triangle angle sum and exterior-angle relationship to solve multi-step problems.*

Find each unknown angle or value.

13. *Triangle sum*

A triangle has angles 65° and 80°. Find the third angle.

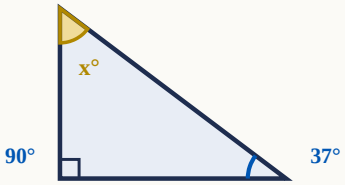


x = _____

answer: 35°

14. *Triangle sum*

A triangle has angles 90° and 37°. Find the third angle.



x = _____

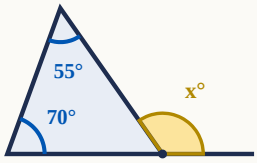
answer: 53°

15. *Exterior angle*

Remote interior angles are 55° and 70°. Find the exterior angle.

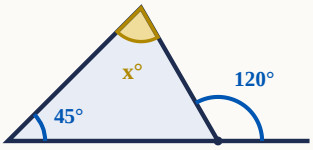
16. *Exterior angle*

An exterior angle is 120°; one remote interior is 45°. Find the other remote.



x = _____

answer: 125°

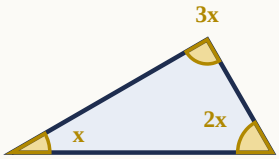


x = _____

answer: 75°

17. *Solve*

A triangle has angles x, 2x, and 3x. Find x.

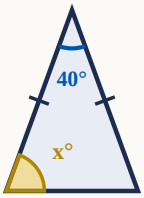


x = _____

answer: x = 30

18. *Isosceles*

An isosceles triangle has a vertex angle of 40°. Find one base angle.



x = _____

answer: 70°